Lab QMS

User Manual

A Guide to Every Module in the Sidebar

What each part does, why it exists, and how it strengthens the laboratory's quality management system under ISO 15189:2022.

Prepared for Medilixmaldives Lab QMS

Contents

Introduction

This manual explains every item in the Lab QMS sidebar — what it does, why it exists, and how using it well strengthens the laboratory's compliance with ISO 15189:2022. It is written for anyone using the system day to day: technologists logging results, a QA Manager reviewing compliance, or an Admin managing staff access.

Each section follows the same structure: how the module works in the app, why that function is required (with the relevant ISO 15189:2022 clause), how it concretely improves the quality management system, and a worked example grounded in real laboratory practice.

A theme runs through all eleven modules: a quality management system is only as strong as its weakest link between finding a problem and proving it was fixed. Every module in this app exists to shorten and strengthen that link — from a Non-conformity being raised, to a task being assigned, to that task's completion being visible on the Dashboard, to the whole chain being retrievable later in the Audit log.

1. Dashboard

The Dashboard is the first screen after signing in. It pulls a small number of the most consequential figures from every other module into one view, so the overall health of the laboratory's quality system is visible at a glance, without opening each register individually.

How it works in the app

- Overall clause compliance percentage, and a breakdown of clause statuses (Compliant, Partial, Non-conformant, Not assessed) across all five clause groups.
- Counts of open tasks and how many are overdue.
- Counts of open non-conformities and how many are Critical severity.
- Staff competency records that are overdue or due within 30 days.
- Equipment IQ/OQ/PQ, calibration, and maintenance records that are overdue or due soon.
- Unauthorized IQC results that carry a Westgard reject-level violation.
- EQA results classified Unsatisfactory.

Why it's needed

A quality system generates records continuously — clause reviews, IQC results, competency checks, equipment maintenance. Without a consolidated view, a problem can exist in the system for weeks before anyone happens to open the specific page where it is visible. Most accreditation bodies expect a laboratory to demonstrate ongoing, active monitoring of its own performance, not periodic rediscovery of problems during audit preparation.

How it helps improve the QMS

The Dashboard turns the QMS from a set of separate registers into a single management instrument. It supports the laboratory director or QA Manager in doing what Clause 8.9 (Management reviews) and Clause 8.8 (Evaluations) both expect: reviewing the system's performance at defined intervals using real, current data, rather than reconstructing a picture from memory or from scattered spreadsheets right before an assessment.

> **Example:** *On a Monday morning, the QA Manager opens the Dashboard and sees "3 unauthorized IQC rule violations" highlighted in red. Investigating immediately, she finds a Westgard 2_2s violation on yesterday's Biochemistry Level 2 control that was never reviewed. Without the Dashboard surfacing this automatically, it might not have been noticed until the next EQA cycle — or until a clinician queried a result.*

Reference: *Supports Clause 8.9 (Management reviews) and Clause 8.8 (Evaluations).*

2. Clause Register

The Clause register is a checklist of all 34 sub-clauses of ISO 15189:2022, organized under the standard's five main clause groups: General requirements (4), Structural and governance requirements (5), Resource requirements (6), Process requirements (7), and Management system requirements (8). Each sub-clause carries a status, an owner, a last-reviewed date, and free-text notes.

How it works in the app

- Every sub-clause can be set to Not assessed, Compliant, Partial, or Non-conformant.
- An owner (a specific staff member) can be assigned to each clause, making one person accountable for it.
- Notes capture evidence references — for example, which SOP or record demonstrates conformity.
- An "Assign task" button on each clause creates a task directly linked to that clause, so a gap becomes tracked work without leaving the page.

Why it's needed

ISO 15189:2022 is organized as 34 distinct sub-clauses, each carrying its own specific requirements. A laboratory cannot credibly claim conformity to "the standard" in the abstract; it must be able to show, clause by clause, how each requirement is being met. Without a structured register, this evidence tends to live in the heads of a few senior staff, or gets reconstructed under time pressure before an assessment visit.

How it helps improve the QMS

The register turns accreditation readiness from an annual event into a continuously maintained state. Because every clause has a named owner, responsibility for each requirement is explicit rather than assumed. Because tasks can be spawned directly from a clause, a finding of "Partial" compliance does not just sit as a status label — it becomes an assigned, due-dated piece of work that shows up elsewhere in the system until it is resolved.

> **Example:** *During a quarterly internal review, Clause 6.2 (Facilities and environmental conditions) is marked Partial because continuous temperature monitoring is missing on cold cabinet 3. The QA Manager clicks "Assign task" directly from that clause row, creating "Install continuous temperature logger — Cabinet 3", due in two weeks, assigned to the facilities lead. The clause stays visibly Partial on the Dashboard until the task is done and the clause is re-reviewed.*

Reference: *ISO 15189:2022, Clauses 4 through 8 in full; the register itself is the primary evidence base for Clause 8.9 (Management reviews).*

3. Tasks

Tasks are assignable, due-dated pieces of work — the mechanism by which a finding anywhere else in the system (a clause gap, an NC, an overdue calibration) becomes a specific person's responsibility with a deadline.

How it works in the app

- Only Admins, Deputy Admins, QA Managers, and Deputy QA Managers can create, reassign, or delete tasks.
- Any signed-in staff member (other than a Viewer) can update the status of a task assigned to them — Open, In progress, or Done — without needing assigner-level access.
- Tasks can optionally link to a specific ISO clause, and can be created directly from the Clause register.
- Overdue tasks are flagged in red and surfaced on the Dashboard.

Why it's needed

Identifying a problem is the easy part of quality management; making sure it actually gets fixed, by a specific person, by a specific date, is where systems tend to fail. Without a task mechanism, follow-up action is usually tracked informally — a verbal request, a sticky note, an email that gets buried — none of which is verifiable evidence for an assessor asking "how do you ensure corrective actions are completed?"

How it helps improve the QMS

Tasks close the loop between finding and fixing. They also enforce a sensible separation of duties: the people responsible for planning and assigning quality work (Admins, QA Managers) are distinct from the broader staff who simply need to report progress on the work they've been given, which keeps the assignment process controlled without making status updates bureaucratic for the technologist doing the work.

> **Example:** *An NC is raised after a delayed specimen turnaround is traced to a broken centrifuge. A task, "Arrange emergency service call for Centrifuge 2", is created and assigned to the equipment coordinator, due the same day. The technologist assigned to receiving deliveries marks a separate task, "Confirm loaner centrifuge received", as Done once it arrives — without needing any elevated permissions to do so.*

Reference: *Supports Clause 8.5 (Actions to address risks and opportunities for improvement) and Clause 8.6 (Improvement).*

4. NC / CAPA

The NC/CAPA register is the full lifecycle record for non-conformities: what went wrong, why, what was done about it, and proof that the fix actually worked before the record is closed.

How it works in the app

- Each NC captures a description, source (e.g. internal audit, EQA/PT failure, complaint, equipment issue), severity (Minor/Major/Critical), and an optional link to a clause.
- A six-stage status path — Open, Investigating, Action planned, Action implemented, Verified, Closed — makes the current stage of remediation explicit.
- Dedicated fields separate root cause, corrective action, and preventive action, so the record distinguishes "what fixed this instance" from "what stops it recurring."
- A verified-by field and closed date ensure someone other than the person who did the fix confirms it worked before the NC is closed.

Why it's needed

Clause 8.7 requires a laboratory to react to nonconformities, evaluate whether corrective action is needed, determine root causes, and verify the effectiveness of whatever action is taken. A verbal "we fixed it" does not satisfy this — the standard expects a structured, evidenced process, and expects preventive action to be considered separately from the immediate fix.

How it helps improve the QMS

Structuring the register this way stops corrective action from being informal symptom-patching. Requiring a documented root cause pushes investigation past the obvious surface cause. Requiring a separate verification step before closure means nobody can mark an NC resolved without someone actually checking the fix held — a common gap in laboratories that treat CAPA as paperwork rather than as a control.

> **Example:** *An EQA cycle returns an unsatisfactory Glucose result. NC-014 is raised with source "EQA/PT failure" and severity Major. Investigation finds a miscalibrated pipette as the root cause. Corrective action: the pipette is recalibrated and the technologist retrained. Preventive action: pipette calibration is added to the routine equipment*

> *maintenance schedule so this class of error is caught earlier next time. The QA Manager verifies the fix against the following EQA cycle's result before closing NC-014.*

Reference: *Clause 8.7 (Nonconformities and corrective actions); also draws on Clause 7.5 (Nonconforming work) for examination-related non-conformities.*

5. IQC & Levey-Jennings

This module manages internal quality control across Hematology, Biochemistry, and Immunochemistry analysers: setting up control levels, logging daily results, automatically evaluating them against the Westgard multirule system, charting them on a Levey-Jennings chart, and requiring a second person's authorization before a result is considered signed off.

How it works in the app

- Analysers are grouped by discipline; each has its own parameters (analytes) and control levels (Level 1/2/3, each with a target mean, SD, and lot number).
- Daily results can be entered individually, through a combined "Daily IQC entry" worksheet covering a whole machine at once, or bulk-imported from a CSV file spanning multiple parameters and machines.
- Every result is automatically evaluated against six Westgard rules — 1_2s (warning), 1_3s, 2_2s, R_4s, 4_1s, and 10x — the moment it is entered, with violations shown directly on the Levey-Jennings chart.
- A reject-level violation cannot be authorized by the technologist who entered it; only an Admin or QA Manager can authorize a result, and the authorizer's initials and timestamp are permanently recorded against that result.

Why it's needed

Clause 7.3.7 requires the laboratory to design an internal quality control system that verifies the achievement of the intended quality of results, to define actions when control results fall outside acceptable limits, and to review IQC data at defined intervals. Manually plotting Levey-Jennings charts and checking multirule logic by eye is slow, and it is easy for a subtle pattern — like two consecutive points drifting the same direction — to be missed when reviewed in isolation rather than in context.

How it helps improve the QMS

Automating the rule evaluation means a violation is caught the moment it happens, not at the next periodic review. Requiring independent authorization builds a second check into the process before an out-of-control run can be treated as valid — directly enforcing the separation between "generating a result" and "approving a result" that most IQC failures in practice trace back to not having.

> **Example:** *A Hematology technologist logs today's Level 2 WBC control result. It sits 2.6 SD above the target mean, and combined with yesterday's result — also more than 2 SD above the mean, on the same side — the system flags a 2_2s rule violation. The technologist cannot authorize this result themselves; it stays flagged as "Awaiting QA Manager sign-off" until the QA Manager reviews the run and the analyser is investigated before further patient samples are released on it.*

Reference: *Clause 7.3.7 (Quality control), a sub-clause of 7.3 Examination processes. The specific rule set applied (1_3s, 2_2s, R_4s, 4_1s, 10x) is the Westgard multirule system, a standard reference framework for clinical laboratory IQC interpretation (see CLSI guideline C24).*

6. EQAS

EQAS records External Quality Assessment (proficiency testing) results by discipline, provider, and cycle, comparing the laboratory's own result against the peer group mean and standard deviation to automatically compute a Standard Deviation Index (SDI) and classify the result.

How it works in the app

- Each entry captures discipline, parameter, provider/scheme, cycle, the laboratory's result, and the peer group mean and SD.
- SDI is calculated automatically as (lab result − peer mean) ÷ peer SD.
- Results are classified Satisfactory, Marginal, or Unsatisfactory based on the SDI, with Unsatisfactory results surfaced on the Dashboard.
- Results can optionally be linked to a specific analyser, so a recurring problem can be traced to a specific machine rather than the discipline in general.

Why it's needed

IQC alone cannot detect a systematic bias against the true value — a control that consistently reads within its own established range will pass IQC even if that range itself has drifted. Clause 7.3.7.3 requires participation in interlaboratory comparison programs specifically to catch this kind of error, by checking the laboratory's results against an external, independent reference.

How it helps improve the QMS

Automating the SDI calculation removes a manual step that is easy to get wrong or skip when reviewing results quickly, reducing the chance a Marginal or Unsatisfactory result is misclassified as acceptable. Keeping a running history by analyte, rather than reviewing each EQA cycle in isolation, makes a recurring pattern — the same analyte drifting Marginal cycle after cycle — visible in a way that a single cycle's result never would be.

> **Example:** *A Biochemistry Urea result returns with an SDI of 2.4, auto-classified Marginal. On its own this might be dismissed as normal variation. Reviewing the EQAS history shows this is the third Marginal Urea result in five cycles — a pattern that prompts a method review rather than being written off individually each time.*

Reference: *Clause 7.3.7.3 (External quality assessment), a sub-clause of 7.3.7 Quality control.*

7. Staff Competency

This module tracks staff training, induction, and competency assessments — initial and ongoing — with the assessment method, assessor, result, and next-due date, so competence is monitored on a schedule rather than assumed indefinitely from a hiring decision.

How it works in the app

- Records cover Initial competency assessment, Ongoing/annual competency assessment, Training, Certification/licence, and Induction.
- Each record notes the assessment method (e.g. direct observation, review of records, result correlation, written/practical test), the assessor, and the result.
- A due date on each record drives an overdue/due-soon indicator that surfaces on the Dashboard.

Why it's needed

Clause 6.1 requires the laboratory to determine the competence needed for each role and to assess personnel against those requirements — explicitly including reassessment at defined intervals, not solely at the point of hire or initial training. Competence that is never reassessed is a common, quietly accumulating accreditation gap: a technologist trained three years ago on a since-updated procedure may never have been re-checked against the current version.

How it helps improve the QMS

Attaching a due date to every competency record converts reassessment from something remembered only when an assessor asks for evidence into an operational habit, since it is visible on the Dashboard well before it becomes overdue. Recording the specific method used for each assessment also gives the laboratory a defensible answer to "how do you know this person is competent?" beyond a general assertion.

> **Example:** *A technologist's annual competency reassessment for manual differential counting is due in three weeks. It appears on the Dashboard's competency-overdue widget with enough lead time for the QA Manager to schedule a direct observation session, rather than the lapse only being noticed when an assessor requests evidence of current competence during a surveillance visit.*

Reference: *Clause 6.1 (Personnel), specifically its ongoing competence assessment requirements.*

8. Equipment Records

Equipment records combine an inventory of laboratory equipment with the full qualification and maintenance history for each item — Installation Qualification (IQ), Operational Qualification (OQ), Performance Qualification (PQ), calibration, and preventive or corrective maintenance.

How it works in the app

- The inventory records each item's category, model, serial number, location, and current status (In service, Out of service, Under qualification, Decommissioned).
- Under each item, a full history of IQ/OQ/PQ, calibration, and maintenance records is kept, each with a date, due date, result, and document reference.
- Overdue or soon-due qualification and calibration records surface on the Dashboard.

Why it's needed

Clause 6.3 requires equipment to be qualified as fit for its intended use before being placed into service, and Clause 6.4 requires equipment to be calibrated at intervals that maintain metrological traceability of results. Both clauses expect this to be an ongoing, documented program — not a one-time check when equipment is first purchased.

How it helps improve the QMS

Because overdue calibration and maintenance items surface automatically on the Dashboard, a lapse is caught before it becomes a compliance finding rather than being discovered when a result looks suspicious or, worse, only during an audit. Keeping the full IQ→OQ→PQ chain against each specific instrument is exactly the traceable qualification history an assessor checks when verifying a piece of equipment is fit for its intended use.

> **Example:** *The Medonic M51's next calibration is due in twelve days. It appears on the Dashboard's equipment-overdue widget with time to schedule the service visit. Without this tracking, the lapse might only be discovered when*

> *a result looks inconsistent — or during the next external audit, by which point an unknown volume of patient results may have been reported using an uncalibrated instrument.*

Reference: *Clause 6.3 (Equipment) and Clause 6.4 (Equipment calibration and metrological traceability).*

9. Documents

Documents is a linked register of the laboratory's controlled documents — SOPs, policies, manuals, calibration certificates, service reports, EQA certificates, and training materials — each recorded with a category, a cross-reference to what it relates to, and a link to where the actual file is stored.

How it works in the app

- Each entry records a title, category, an optional "related to" cross-reference (e.g. an equipment name, a clause number, an NC number), and a link to the source file.
- The register stores links and metadata rather than the files themselves, so the authoritative copy of each document continues to live in the laboratory's existing document storage (e.g. a shared drive).

Why it's needed

Clause 8.3 requires management system documents to be controlled: identifiable, current, accessible to those who need them, and traceable to what they support. Without an index, supporting evidence for a given SOP, calibration, or NC tends to be scattered across email, personal drives, and paper folders, making it difficult for anyone other than the person who filed it to locate.

How it helps improve the QMS

The register gives every other module in the system a consistent place to point to its supporting evidence, rather than evidence existing only as a filename someone remembers. This is particularly valuable when an assessor traces a specific record — an NC, a calibration, a competency assessment — back to the document that supports it, since the cross-reference makes that trail explicit rather than requiring it to be reconstructed on request.

> **Example:** *NC-014 (the Glucose EQA failure described earlier) references "Pipette Calibration Certificate — March 2026" in its evidence field. That certificate is separately logged in Documents under the Calibration certificate category, cross-referenced to the equipment item, so anyone tracing NC-014 back to its supporting evidence finds the document indexed rather than having to ask where the file is kept.*

Reference: *Clause 8.3 (Control of management system documents).*

10. Personnel

Personnel is the staff directory that underlies every other module — every "assigned to," "operator," "owner," "authorized by," and "raised by" field elsewhere in the system draws on this list. It is also where staff logins and access levels are managed.

How it works in the app

- Each person's name, job title, email, and record card number (their login username) are recorded here.
- Admins create new staff logins and set each person's access role — Admin, Deputy Admin, QA Manager, Deputy QA Manager, Technologist, or Viewer — at the point of creation.
- Access role is auto-detected from the account at login; staff never choose their own access level, which prevents self-escalation.
- Admins can update an existing person's record card number, access role, and password.

Why it's needed

Clause 6.1 requires records of relevant personnel, including their authority and responsibilities. Separately, the standard's expectations around structure and authority (Clause 5.4) and impartiality (Clause 4.1) are only meaningfully enforced if the software actually restricts who can perform sensitive actions — such as authorizing an IQC result or closing an NC — to the people who hold that authority, rather than relying on everyone voluntarily following an unenforced policy.

How it helps improve the QMS

Personnel is what makes every attribution elsewhere in the system trustworthy: an "authorized by" field is only meaningful evidence if the software genuinely prevented anyone else from performing that action. Centralizing access control here also means a single change — updating one person's role — correctly and immediately changes what they can do everywhere in the system, rather than requiring access to be managed separately in each module.

> **Example:** *Ahead of the QA Manager's planned leave, an Admin promotes a senior technologist to Deputy QA Manager so IQC authorization coverage continues without interruption. The change itself, including who made it and when, is recorded automatically in the Audit log.*

Reference: *Clause 6.1 (Personnel); Clause 5.4 (Structure and authority).*

11. Audit & Backup

Audit & Backup provides two things: an automatic, database-enforced log of every create, update, and delete made anywhere in the system, and a one-click export of the entire dataset to a single Excel workbook.

How it works in the app

- Every change anywhere in the system is logged automatically, attributed to whoever was signed in when it happened — this is written by the database itself, not by application code staff could forget to trigger.
- The audit log can be filtered by entity type (e.g. "IQC result", "Non-conformity") and by the person who made the change.
- The full backup exports every module — Personnel, Clause status, Tasks, NC/CAPA, Competency, Equipment, IQC, EQA, Documents, and the audit log itself — as one workbook, one sheet per module.

Why it's needed

Clause 8.4 requires records to be protected, retrievable, and retained for a defined period, and expects the laboratory to be able to demonstrate the integrity of those records. Being able to show who changed a specific value, and when, is central to that expectation — and is often specifically requested by accreditation bodies and, in many jurisdictions, health regulators, particularly for records with clinical consequences such as IQC target values or authorized results.

How it helps improve the QMS

Because the log is written automatically by the database rather than by staff remembering to record a change, it cannot be selectively skipped, and it captures the full picture rather than only the changes someone thought to document. The backup export gives the laboratory an independent copy of its own records, which matters regardless of the reliability of whatever platform is hosting the live system — records should not depend entirely on one system staying available indefinitely.

> **Example:** *A QC control's target mean was edited last month, and a result that once looked borderline against the old mean now looks squarely in range. The Audit log shows precisely when the mean was changed, by whom, and what it changed from and to — evidence that would otherwise depend entirely on someone's memory of a change made weeks earlier.*

Reference: *Clause 8.4 (Control of records); the log itself is a natural source document for Clause 8.8.3 (Internal audits).*

How the Modules Work Together

None of these eleven modules is meant to stand alone. The value of the system comes from how a finding in one module becomes tracked, evidenced action in another:

- A gap found in the Clause register becomes a Task, assigned to a named owner with a due date.
- A Task that reveals a deeper problem becomes an NC, with a documented root cause and corrective/preventive action.
- An NC's evidence — a certificate, a report, a corrected SOP — is indexed in Documents so it can be found again.
- IQC and EQAS results that go out of control generate the Non-conformities that feed the same NC/CAPA process.
- Equipment and Staff competency both feed the Dashboard's early-warning widgets, so lapses are caught before they cause an NC in the first place.
- Every single one of these actions — across every module — is captured without extra effort in the Audit log, and can be exported as a complete backup at any time.

Used this way, the system's real output is not any one register, but a continuous, evidenced chain from "a requirement exists" through "a gap was found" to "the gap was fixed, verified, and the proof is retrievable" — which is, in essence, what ISO 15189:2022 asks a laboratory's quality management system to be able to demonstrate.

Quick-Reference: Clause Mapping

For fast lookup when preparing for an internal review or external assessment.

Module	Primary ISO 15189:2022 reference
Dashboard	Clause 8.9 (Management reviews); Clause 8.8 (Evaluations)
Clause register	Clauses 4–8 in full; primary evidence base for Clause 8.9
Tasks	Clause 8.5 (Actions to address risks and opportunities); Clause 8.6 (Improvement)
NC / CAPA	Clause 8.7 (Nonconformities and corrective actions); Clause 7.5 (Nonconforming work)
IQC & Levey-Jennings	Clause 7.3.7 (Quality control)
EQAS	Clause 7.3.7.3 (External quality assessment)
Staff competency	Clause 6.1 (Personnel) — ongoing competence assessment
Equipment records	Clause 6.3 (Equipment); Clause 6.4 (Equipment calibration and metrological traceability)
Documents	Clause 8.3 (Control of management system documents)
Personnel	Clause 6.1 (Personnel); Clause 5.4 (Structure and authority)
Audit & Backup	Clause 8.4 (Control of records); supports Clause 8.8.3 (Internal audits)